

EXTENSION 1: PI v.s. CI

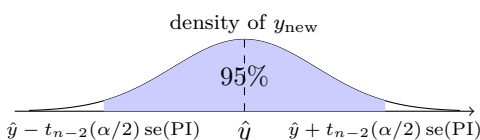
Problem Setup 1. Consider the SLR model $y = \beta_0 + \beta_1 x + \epsilon$, where $\epsilon \sim N(0, \sigma^2)$,

- y is the house price, and x is the house area,
- $(x_i, y_i)_{i=1}^n$ is the observed data, and we already computed $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\sigma}^2$.

Now I want to buy a particular house with area $x_0 = 100$. So we treat it as a new random variable:

$$y_{\text{new}} = \beta_0 + \beta_1 x_0 + \varepsilon_{\text{new}}.$$

To get a prediction for this rv, we use its distribution and draw out a large probability region, specifically

$$\frac{y_{\text{new}} - \hat{y}}{\text{se}(\text{PI})} \sim t_{n-2} \quad \text{where} \quad \text{se}(\text{PI}) = \sqrt{\hat{\sigma}^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}$$


This means the new rv y_{new} can vary, and it has a high probability to lie in the shaded area. So we call this interval $[\hat{y} - t_{n-2}(\alpha/2) \text{se}(\text{PI}), \hat{y} + t_{n-2}(\alpha/2) \text{se}(\text{PI})]$ the *prediction interval* of the individual y_{new} .

Now I want to estimate the mean price for all houses with area $x_0 = 100$. So we are interested in

$$\mathbb{E}[y|x_0] = \beta_0 + \beta_1 x_0 \quad \text{which is NOT rv but an unknown parameter.}$$

So the CI strategy for parameter can apply. Specifically, we find the unbiased estimator $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_0$ for $\mathbb{E}[y|x_0]$ and can derive

$$\frac{\hat{y} - \mathbb{E}[y|x_0]}{\text{se}(\text{CI})} \sim t_{n-2} \quad \text{where} \quad \text{se}(\text{CI}) = \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}.$$

Then we can write

$$\begin{aligned} 1 - \alpha &= \mathbb{P} \left(t_{n-2}(\alpha/2) \leq \frac{\hat{y} - \mathbb{E}[y|x_0]}{\text{se}(\text{CI})} \leq t_{n-2}(\alpha/2) \right) \\ &= \dots \text{for you to complete} \dots \\ &= \mathbb{P} \left\{ \hat{y} - t_{n-2} \left(\frac{\alpha}{2} \right) \text{se}(\text{CI}) \leq \mathbb{E}[y|x_0] \leq \hat{y} + t_{n-2} \left(\frac{\alpha}{2} \right) \text{se}(\text{CI}) \right\}. \end{aligned}$$

Here our estimate goal $\mathbb{E}[y|x_0]$ does NOT vary, but the interval $[\hat{y} - t_{n-2}(\alpha/2) \text{se}(\text{CI}), \hat{y} + t_{n-2}(\alpha/2) \text{se}(\text{CI})]$ varies as we take different samples. But if we take 100 samples to get 100 intervals, roughly 95 of them contains the true $\mathbb{E}[y|x_0]$. So in our one particular CI, we are quite confident that $\mathbb{E}[y|x_0]$ is in it. This gives the name *confidence interval*.

Which one to use in exams/homework? If the question explicitly asks for a CI or a PI, use the required one. Otherwise, use a PI for predicting a new individual response, and a CI for estimating the mean response.

EXTENSION 2: INTERPRETING COEFFICIENTS

Problem Setup 2. Consider the MLR model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$ where $\epsilon \sim N(0, \sigma^2)$,

- y represents the badminton level,
- x_1 represents height, and x_2 represents leg length,
- and the fitted Model 0 is $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 = x_1 + 2x_2$

A naive interpretation of $\hat{\beta}_2 = 2$ is: holding x_1 fixed, a one-unit increase in x_2 raises y by 2. This would be literal only if leg length could change without height. In practice we can never split the increment of them, so we need a different interpretation.

In Question 3 & 4 of Tutorial 3, we fitted 3 models:

$$\left. \begin{array}{l} \text{Model 1: } y \sim x_1 \rightarrow \text{get residuals } e_{y \cdot x_1} \\ \text{Model 2: } x_2 \sim x_1 \rightarrow \text{get residuals } e_{x_2 \cdot x_1} \end{array} \right\} \Rightarrow \text{Model 3: } e_{y \cdot x_1} \sim e_{x_2 \cdot x_1},$$

where we can view $e_{y \cdot x_1}$ as the part of the response unexplained by height, and $e_{x_2 \cdot x_1}$ as the part of leg length unexplained by height x_1 . We find both computation (Que3) and theory (Que4) show that:

- the fitted Model 3 is $\hat{e}_{y \cdot x_1} = 0 + \hat{\beta}_2 e_{x_2 \cdot x_1}$, i.e. Model 3 and Model 0 share the same $\hat{\beta}_2$,

This gives another interpretation of $\hat{\beta}_2$ as follows:

After removing the linear effect of height x_1 from both the response y and the predictor x_2 , the coefficient $\hat{\beta}_2$ measures how the remaining part in leg length is associated with the remaining part in badminton level. In short, $\hat{\beta}_2$ measures the contribution of x_2 on y after both have been linearly adjusted for x_1 .

EXTENSION 3: $\text{sampCor}(x, y) = \sqrt{R^2}$ IN SLR

Proof. We will prove it by squaring both sides, i.e. to prove $[\text{sampCor}(x, y)]^2 = R^2$. Starting with RHS:

$$R^2 = \frac{SS_R}{SS_T} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2},$$

in Tutorial 1, we showed $\sum_{i=1}^n (\hat{y}_i - y_i) = 0$, which means $\bar{y}$ is also the sample mean of $\hat{y} := (\hat{y}_1, \dots, \hat{y}_n)$. So, the numerator $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = (n-1) \times \text{sampVar}(\hat{y})$. And the denominator is just our definition S_{yy} . So,

$$= \frac{(n-1) \cdot \text{sampVar}(\hat{y})}{S_{yy}} = \frac{(n-1) \cdot \text{sampVar}(\hat{\beta}_0 + \hat{\beta}_1 x)}{S_{yy}}$$

$\hat{\beta}_0, \hat{\beta}_1$ are some computed numbers, say 1 or 2, we can take them out and get

$$= \frac{(n-1) \hat{\beta}_1^2 \cdot \text{sampVar}(x)}{S_{yy}}$$

combining $(n-1)$ and $\text{sampVar}(x)$, we get

$$= \frac{S_{xx}}{S_{yy}} \times \hat{\beta}_1^2 = \frac{S_{xx}}{S_{yy}} \times \left(\frac{S_{xy}}{S_{xx}} \right)^2 = \frac{S_{xy}^2}{S_{xx} S_{yy}} = [\text{sampCor}(x, y)]^2.$$